

Chia-Chien (Aaron) Hung

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

expected May 2027

Bachelor of Science in Computer Science, Concentration in Intelligence and Modeling/Simulation

- GPA: 4.0/4.0 (Faculty Honors)
- Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Computer Organization and Programming, Systems and Networks, Discrete Mathematics, Multivariable Calculus, Differential Equations

Bachelor of Science in Biochemistry, Pre-Health Concentration

- GPA: 4.0/4.0 (Faculty Honors)
- Relevant Coursework: General Chemistry, Organic Chemistry, Synthesis Laboratory

RESEARCH

Student Researcher

Atlanta, GA

SIPLab, Georgia Tech

May 2024 - Present

- Engineered a model predictive controller in C/C++ for high-performance computing, achieving a 25-fold improvement in runtime.
- Co-authoring a forthcoming research paper that evaluates diverse control algorithms for neuroscience applications.
- Implementing and training deep learning control algorithms to maintain neuron membrane potential near target range.

Student Researcher

Atlanta, GA

Integrative Systems Biology Lab, Georgia Tech

December 2024 - Present

- Developing an advanced computational model to simulate dynein-driven cell metastasis.
- Leveraging optimization and parallelization techniques to accelerate large-scale simulation runtimes on high-performance computing clusters.

Research Assistant

Atlanta, GA

Coskun Lab, Georgia Tech

December 2024 - Present

- Acquiring and applying wet-lab techniques in immunofluorescence staining and imaging.
- Collaborating with master's students to refine imaging protocols and develop assays to detect markers for chronic kidney transplant rejection.

Cancer Research Intern

Taipei, Taiwan

Institute of Statistical Science, Academia Sinica

March 2021 - August 2023

- Conducted comprehensive research into the theoretical foundations of Molecular Biology, Cancer Biology, and Statistics.
- Developed an integrated analytical pipeline to assess viral contributions to lung cancer, utilizing advanced statistical methods for data processing.
- Earned an Honorable Mention for research presentation at the 2023 Taiwan-Japan Joint Conference hosted by the Taiwan Genomics and Genetics Society.

EXPERIENCE

Software Engineering Lead

Atlanta, GA

Molecular Evolution Core Lab, Georgia Tech

January 2024 - Present

- Leading a 15-member software engineering team to design and implement a laboratory information management system using Node.js, React.js, and MongoDB, thereby enhancing sample tracking efficiency and accuracy.
- Architected and deployed robust APIs while integrating third-party services to extend application functionality and streamline workflows.
- Facilitated regular code reviews and feedback sessions to foster a collaborative and high-performance team culture.

SKILLS AND COMPETENCIES

Development Tools: Git, OpenCV, CMake, Visual Studio Code, Android Studio, Agile, Linux, Arduino

Programming Languages: C/C++, Java, Python, JavaScript, TypeScript, HTML, CSS, React, LaTeX, Assembly